Appendix III

Task Plan of Thesis Project

Project Title: *Medical Reasoning with Memory-Augmented Agentic System*

Progress Timeline: (From Oct 18th 2025 to May 22nd 2026)

Milestone	Date	Description
Project Initialization	Oct 18 - Nov 15, 2025	Confirm thesis topic and initial project scope definition.
Literature Review	Nov 16, 2025 - Jan 7, 2026	Complete a comprehensive review of medical datasets, LLM-based agents, and memory-augmented frameworks.
Proposal Preparation	Jan 8 - Jan 23, 2026	Prepare the thesis proposal.
Prototype Development	Jan 24 - Mar 20, 2026	Design the modular architecture integrating dynamic memory retrieval, case-based reasoning, and LLM inference. Establish testing environment.
Mid-Term Check-Up	Mar 27, 2026	Demonstrate preliminary prototype and discuss methodology improvements.
Experimental Implementation & Analysis	Mar 28 – May 10, 2026	Conduct experiments on benchmark datasets, evaluating results and refining architecture.
Thesis Draft Preparation	May 11 – May 18, 2026	Compile results and finalize the full thesis draft for review.

Task details of thesis project:

Comprehensive Literature Review
Review research on memory-augmented reasoning systems and clinical decision-support frameworks; identify existing limitations in reasoning continuity and interpretability.

Prototype Development
Build a functional prototype implementing core reasoning loops and memory components, integrating large language models with dynamic memory retrieval and case-based reasoning modules to simulate clinical diagnostic reasoning.

Experimental Evaluation
Design experiments to test the system's diagnostic reasoning accuracy, temporal coherence, and explanation quality compared with baseline LLM approaches.

Supervisor Name & Signature:	**Zuozhu Liu** 刘佐炷	**Date:** **Oct 31,2025**